

Algorithm for finding A^{-1}

Row reduce the augmented matrix $[A \ I]$. If A is row equivalent to I , then $[A \ I]$ is row equivalent to $[I \ A^{-1}]$. Otherwise, A does not have an inverse.

Example 7: Find the inverse of the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$, if it exists.

Solution: $[A \ I] = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix}$ Row 1 and Row 2 are interchanged

$$\begin{array}{l} \text{Row 3} = \text{Row 3} - 4 \times \text{Row 2} \\ \text{Row 3} = \text{Row 3} - 4 \times \text{Row 1} \end{array} \rightarrow \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{bmatrix} \xrightarrow{\text{Row 3} = \text{Row 3} + 3 \times \text{Row 2}} \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{bmatrix}$$

$$\begin{array}{l} \text{Row 3} = \text{Row 3} \times 1/2 \\ \text{Row 1} = \text{Row 1} - 3 \times \text{Row 3} \end{array} \rightarrow \begin{bmatrix} 1 & 0 & 3/2 & 0 & 1 & -3/2 \\ 0 & 1 & 1 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix} \xrightarrow{\text{Row 1} = \text{Row 1} - 3 \times \text{Row 3}} \begin{bmatrix} 1 & 0 & 0 & -9/2 & 7 & -3/2 \\ 0 & 1 & 1 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix}$$

$$\begin{array}{l} \text{Row 2} = \text{Row 2} - \text{Row 3} \\ \text{Row 2} = \text{Row 2} \times 1/2 \end{array} \rightarrow \begin{bmatrix} 1 & 0 & 0 & -9/2 & 7 & -3/2 \\ 0 & 1 & 0 & -1/2 & 2 & -1/2 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix}$$

$$\text{By Theorem 7, } A^{-1} = \begin{bmatrix} -9/2 & 7 & -3/2 \\ -1/2 & 2 & -1/2 \\ 3/2 & -2 & 1/2 \end{bmatrix}$$

It is a good idea to check the final answer:

$$AA^{-1} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix} \begin{bmatrix} -9/2 & 7 & -3/2 \\ -1/2 & 2 & -1/2 \\ 3/2 & -2 & 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \cdot (-9/2) + 1 \cdot (-1/2) + 2 \cdot (3/2) & 0 \cdot 7 + 1 \cdot 2 + 2 \cdot (-2) & 0 \cdot (-3/2) + 1 \cdot (-1/2) + 2 \cdot (1/2) \\ 1 \cdot (-9/2) + 0 \cdot (-1/2) + 3 \cdot (3/2) & 1 \cdot 7 + 0 \cdot 2 + 3 \cdot (-2) & 1 \cdot (-3/2) + 0 \cdot (-1/2) + 3 \cdot (1/2) \\ 4 \cdot (-9/2) + (-3) \cdot (-1/2) + 8 \cdot (3/2) & 4 \cdot 7 + (-3) \cdot 2 + 8 \cdot (-2) & 4 \cdot (-3/2) + (-3) \cdot (-1/2) + 8 \cdot (1/2) \end{bmatrix}$$